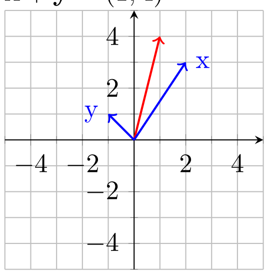
MTH 336(A) Dr. James Ashe

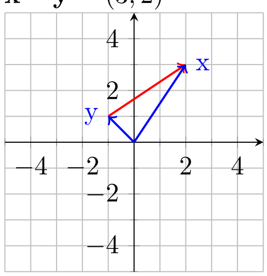
HW 1.1 Solutions Monday 26^{th} August, 2013 **Name:**

1. $\mathbf{x} = (2,3)$ and $\mathbf{y} = (-1,1)$

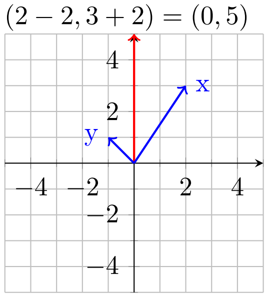
a. $\mathbf{x}+\mathbf{y} = (1,4)$



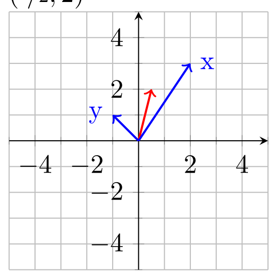
b. $\mathbf{x}-\mathbf{y} = (3,2)$



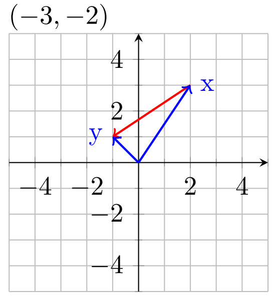
c. $\mathbf{x}+2\mathbf{y} = (2,3)+2(-1,1) = (2-2, 3+2) = (0,5)$



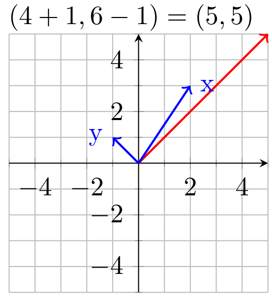
d. $^1/_2\mathbf{x} + {^1/_2}\mathbf{y} = {^1/_2}(\mathbf{x}+\mathbf{y}) = ({^1/_2}, 2)$



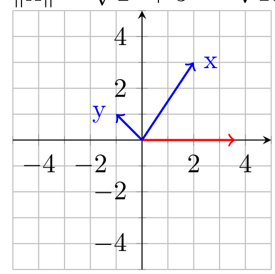
e. $\mathbf{y}-\mathbf{x} = -(\mathbf{x}-\mathbf{y}) = (-3,-2)$



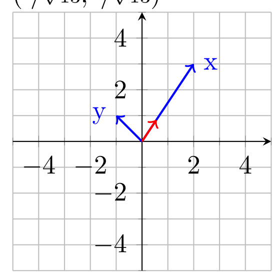
f. $2\mathbf{x}-\mathbf{y} = 2(2,3)-(-1,1) = (4+1, 6-1) = (5,5)$



g. $\|\mathbf{x}\| = \sqrt{2^2+3^2} = \sqrt{13}$



h. $^{\mathbf{x}}/_{\|\mathbf{x}\|} = {^1/_{\sqrt{13}}}(2,3) = ({^2/_{\sqrt{13}}}, {^3/_{\sqrt{13}}})$



This is the *unit vector* as $\|^{\mathbf{x}}/_{\|\mathbf{x}\|}\| = 1$.

5. If a point (x_1, x_2) is on the line ℓ then $x_1 = 1-2t$ and $x_2 = 3+t$ for some $t \in \mathbb{R}$.

- **a.** $x_1 = -1 \Rightarrow t = 1$ so $x_2 = 3+1 = 4$. This point lies on ℓ.
- **b.** $x_2 = 0 \Rightarrow t = -3$ so $x_1 = 1-2(-3) = 7$. This point lies on ℓ.
- **c.** $x_2 = 2 \Rightarrow t = -1$ so $x_1 = 3$. This point does not lie on ℓ.

6. (There may be multiple solutions)

- **a.** If $3x_1 + 4x_2 = 6$ then $x_1 = -{^4/_3}x_2 + 2$. Thus $\mathbf{x} = (x_1, x_2) = (-{^4/_3}x_2 + 2, x_2) = x_2(-{^4/_3}, 1) + (2,0)$. Replacing $x_2 = t$, $\mathbf{x} = t(-{^4/_3}, 1) + (2,0)$
- **b.** The slope $^1/_3$ gives the direction vector $(3,1)$ for a line that starts at $(-1,2)$, $\mathbf{x} = (-1,2) + t(3,1)$.
- **d.** The direction vector is $(3,5)$ for a line that starts at $(-2,1)$, $\mathbf{x} = (-2,1) + t(3,5)$.

10.a. The plane containing the point $(-1,0,1)$ and the line $\mathbf{x} = (1,1,1) + t(1,7,-1)$ also contains the point $(1,1,1)$ (just let $t=0$) and is parallel to the vector $(1,7,-1)$.

- We need two non-parallel lines which are both parallel to the plane. To find another vector note that the difference between the vectors, $(1,1,1)-(-1,0,1) = (2,1,0)$, must be parallel to the plane since both these points are on the plane.

Now $\mathbf{x} = (1,1,1) + t(1,7,-1) + s(2,1,0)$, for $s,t \in \mathbb{R}$.

13. $\overrightarrow{AM} = \frac{1}{2}\overrightarrow{AB}$ and $\overrightarrow{AN} = \frac{1}{2}\overrightarrow{AC}$.

So then $\overrightarrow{MN} = \overrightarrow{AN} - \overrightarrow{AM} = \frac{1}{2}\overrightarrow{AC} - \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\left(\overrightarrow{AC} - \overrightarrow{AB}\right) = \frac{1}{2}\overrightarrow{BC}$.

22. **a.** $\mathbf{v} = c_1\mathbf{v_1} + c_2\mathbf{v_2} + \cdots + c_2\mathbf{v_k}$ where $c_i \in \mathbb{R}$ for each $i = 1, \ldots, k$. So then for any $c \in \mathbb{R}$, $c\mathbf{v} = cc_1\mathbf{v_1} + cc_2\mathbf{v_2} + \cdots + cc_k\mathbf{v_k}$ is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$ since $cc_i \in \mathbb{R}$ for each $i = 1, \ldots, k$.

b. $\mathbf{v} = c_1\mathbf{v_1} + c_2\mathbf{v_2} + \cdots + c_2\mathbf{v_k}$ and $\mathbf{w} = d_1\mathbf{v_1} + d_2\mathbf{v_2} + \cdots + d_k\mathbf{v_k}$ where $c_i, d_i \in \mathbb{R}$ for each $i = 1, \ldots, k$. So then

$$
\begin{aligned}
\mathbf{v} + \mathbf{w} &= c_1\mathbf{v_1} + c_2\mathbf{v_2} + \cdots + c_k\mathbf{v_k} + d_1\mathbf{v_1} + d_2\mathbf{v_2} + \cdots + d_k\mathbf{v_k} \\
&= c_1\mathbf{v_1} + d_1\mathbf{v_1} + c_2\mathbf{v_2} + d_2\mathbf{v_2} + \cdots + d_k\mathbf{v_k} \cdots + d_k\mathbf{v_k} \\
&= (c_1 + d_1)\mathbf{v_1} + (c_2 + d_2)\mathbf{v_2} + \cdots + (c_k + d_k)\mathbf{v_k}
\end{aligned}
$$

Which is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$.